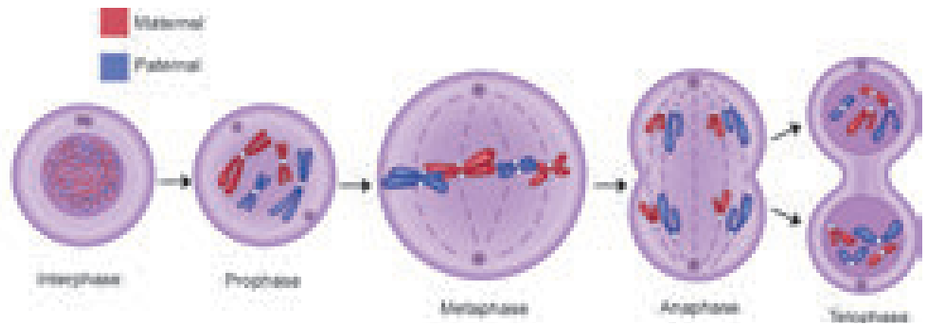


Figure 18.22 Events of Mitosis

same number of chromosomes and genes as the parent cells. It is also called as **equational division**. Mitosis consists of two events, they are: 1. Karyokinesis 2. Cytokinesis

Interphase: It is the **resting phase** of the nucleus. It is the interval between two successive cell divisions. The cell prepares itself for the next cell division.

1. Karyokinesis

The division of the nucleus into two daughter nuclei is called Karyokinesis. It consists of four phases. They are: Prophase, Metaphase, Anaphase and Telophase.

Prophase (pro-first): During this stage chromosomes become short and thick and are clearly visible inside the nucleus. Centrosome splits into centrioles and occupy opposite poles of the cell. Each centriole is surrounded by **aster rays**. Spindle fibres appear between the two centrioles. Nuclear membrane and nucleolus disappear gradually.

Metaphase (meta – after): The duplicated chromosomes arrange on the equatorial plane and form the metaphase plate. Each chromosome gets attached to a spindle fibre by its centromere. The centromere of each chromosome divides into two each being associated with a chromatid.

Anaphase (ana – up, back): The centromeres attaching the two chromatids divide and the two daughter chromatids of each chromosome separate and migrate towards the two opposite poles.

Telophase (tele – end): Each chromatid (or) daughter chromosome lengthens, becomes thinner and turns into a network of chromatin threads. Spindle fibres breakdown and disappear. Nuclear membrane and nucleolus reappear in each daughter nucleus.

2. Cytokinesis

The division of the cytoplasm into two daughter cells by constriction of the cell membrane is called cytokinesis.

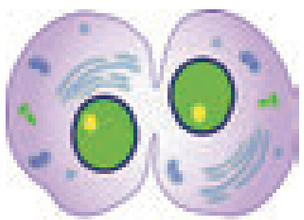


Figure 18.23 Cytokinesis

Significance of Mitosis

1. This equational division results in the production of diploid daughter cells (2n) with equal distribution of genetic material (DNA).
2. In multicellular organisms growth, organ development and increase in body size are accomplished through the process of mitosis
3. Mitosis helps in repair of damaged and wounded tissues by renewal of the lost cells.

III. Meiosis

The term meiosis was coined by Farmer in 1905. It is the kind of cell division that produces the sex cells or the gametes. It is also called reduction division because the chromosome number is reduced to haploid (n) from diploid (2n). Meiosis produces four daughter cells from a parent cell.

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